Ledergor 2018 multiple myeloma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

16/18 samples (≥183 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 13 detached; iLISI median 1.00→5.13 of 16 ≈ 32% of ideal; kNN graph = 1 component at every θ). Self-neighbour enrichment falls with integration — Harmony works, not an archipelago.

Gate 2 — module geography → GO (non-branching)

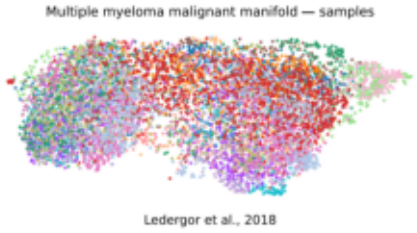
Single dominant pSom→Crest axis: pSom high at the Gast/pSom-high origin declines along pseudotime (ρ=−0.56); Crest anti-correlated (smoothed r=−0.96) and rises to a terminal readout (ρ=+0.57; crest-high decile mean pt=0.81). Gast weak/flat. Axis holds within patients (r=−0.64).

Gate 3 — cross-repression → NA

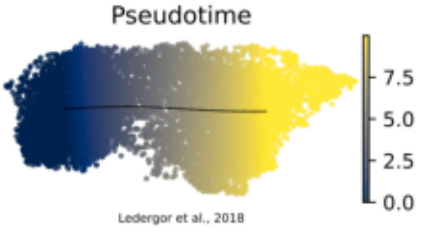
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test.

STEP FIGURES

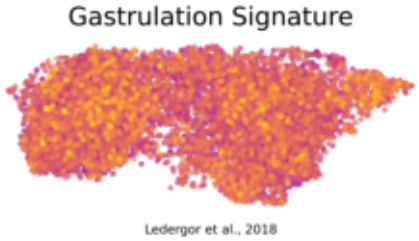
Manifold — samples



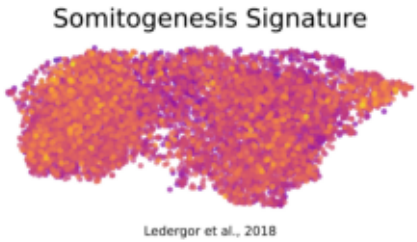
Pseudotime



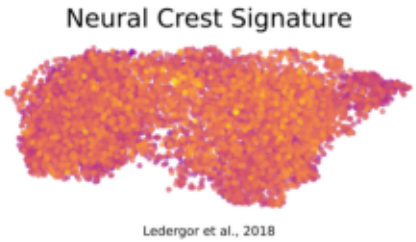
Gastrulation



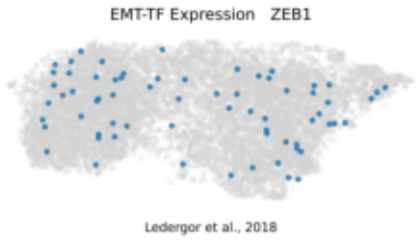
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma (Ledergor et al., 2018), MARS-seq — 8,273 malignant plasma cells load (MM+SMM marrow); 8,197 across 16 samples (≥183) enter preprocessing; 8,184 after island removal; the whole malignant manifold (no lobe scoping) enters the trajectory. Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The myeloma malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis is a within-patient cell-state gradient (77–86% of module variance within-sample), not clonal/CNV structure, and is orthogonal to cell cycle (0.3% cycling). EMT-TFs: only ZEB1 (3.3%) / ZEB2 (1.4%) clear the detection floor — MARS-seq dropout silences the panel.

Parameter choices

Min cells/sample	≥183 (16/18; gap 183→50)	**Integration**	Harmony θ=20; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root	tip: Gast/pSom-high, Crest-low	**Geometry**	Linear (pSom/Crest); 0 stable forks
Island removal	13 cells (radius-graph)	**EMT-TFs**	ZEB1 (+ZEB2); rest <3% detection (MARS-seq dropout)